1. Cauchy-Euler Differential Equations

In the previous module, we solved Higher-Order Linear ODEs where the coefficients were constants. Now, we will look at a special type of linear equation where the coefficients are variables—specifically, powers of x.

This specific form is known as the **Cauchy-Euler Equation** (or Cauchy's Differential Equation).

---

1. Concept Overview

A Cauchy-Euler equation has a very recognizable signature: **the power of the independent variable x exactly matches the order of the derivative it is multiplied by.**

The Standard Form

$$a_n x^n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 x \frac{dy}{dx} + a_0 y = F(x)$$

Where $a_0, a_1, \dots, a_n$ are constants.

Notice that x^2 is paired with $\frac{d^2y}{dx^2}$, x^1 is paired with $\frac{dy}{dx}$, and so on.

---

2. Working Rule (The $x = e^t$ Substitution)

We cannot solve this directly using our constant-coefficient methods. However, we can use a clever substitution to transform the independent variable from x to t, which magically turns the variable coefficients into constant coefficients!

Step 1: The Core Substitution Let $x = e^t$. This also means that $t = \ln x$.

Step 2: Define the New Operator Let $D = \frac{d}{dt}$ (Notice the derivative is now with respect to t, not x).

Step 3: Transform the Derivatives Using the chain rule (proof omitted for brevity), the x-derivatives transform into D-operators as follows: * $x\frac{dy}{dx} = Dy$ * $x^2\frac{d^2y}{dx^2} = D(D-1)y = (D^2 - D)y$ * $x^3\frac{d^3y}{dx^3} = D(D-1)(D-2)y = (D^3 - 3D^2 + 2D)y$

Step 4: Substitute and Solve Substitute these operators into the original equation. It will now be a standard Higher-Order Linear ODE with constant coefficients. Solve for $y(t) = y_c + y_p$ using the methods from Module 4.

Step 5: Resubstitute back to x Once you have the final answer in terms of t, replace every t with $\ln x$ and every e^{kt} with x^k.

---

3. Solved Questions

Here are two comprehensive examples from the lecture notes demonstrating this transformation.

Question 1: Algebraic Right-Hand Side

Solve: $x^2\frac{d^2y}{dx^2} + 7x\frac{dy}{dx} + 5y = x^5$

Solution:

Step 1: Apply Transformations Let $x = e^t$, $t = \ln x$, and $D = d/dt$. Substitute the operators into the left side, and e^t into the right side:

$$[D(D-1) + 7D + 5]y = (e^t)^5$$

$$(D^2 - D + 7D + 5)y = e^{5t}$$

$$(D^2 + 6D + 5)y = e^{5t}$$

(This is now a constant-coefficient equation!)

Step 2: Find the Complementary Function (y_c) Auxiliary Equation: $m^2 + 6m + 5 = 0$ Factor: $(m+5)(m+1) = 0 \implies m = -1, -5$

$$y_c = C_1 e^{-t} + C_2 e^{-5t}$$

Step 3: Find the Particular Integral (y_p)

$$y_p = \frac{1}{D^2 + 6D + 5} e^{5t}$$

Apply the Exponential Rule ($D \to 5$):

$$y_p = \frac{1}{5^2 + 6(5) + 5} e^{5t} = \frac{1}{25 + 30 + 5} e^{5t} = \frac{1}{60} e^{5t}$$

Step 4: Write General Solution in t

$$y(t) = y_c + y_p = C_1 e^{-t} + C_2 e^{-5t} + \frac{1}{60} e^{5t}$$

Step 5: Resubstitute back to x Recall that $e^t = x$, so $e^{-t} = x^{-1}$ and $e^{5t} = x^5$.

$$y(x) = C_1 x^{-1} + C_2 x^{-5} + \frac{x^5}{60}$$

Question 2: Trigonometric & Shift Rule Combination

Solve: $x^2 \frac{d^2y}{dx^2} - 3x \frac{dy}{dx} + 5y = x^2 \sin(\ln x)$

Solution:

Step 1: Apply Transformations Let $x = e^t$, $t = \ln x$, and $D = d/dt$. Substitute into the equation:

$$[D(D-1) - 3D + 5]y = (e^t)^2 \sin(t)$$

$$(D^2 - D - 3D + 5)y = e^{2t} \sin t$$

$$(D^2 - 4D + 5)y = e^{2t} \sin t$$

Step 2: Find the Complementary Function (y_c) Auxiliary Equation: $m^2 - 4m + 5 = 0$ Use quadratic formula: $m = \frac{4 \pm \sqrt{16-20}}{2} = \frac{4 \pm 2i}{2} = 2 \pm i$ Complex roots ($\alpha = 2, \beta = 1$):

$$y_c = e^{2t}(C_1 \cos t + C_2 \sin t)$$

Step 3: Find the Particular Integral (y_p)

$$y_p = \frac{1}{D^2 - 4D + 5}(e^{2t} \sin t)$$

Apply Exponential Shift Rule (Shift e^{2t} to the front, replace $D \to D+2$):

$$y_p = e^{2t} \frac{1}{(D+2)^2 - 4(D+2) + 5} \sin t$$

$$y_p = e^{2t} \frac{1}{(D^2 + 4D + 4) - 4D - 8 + 5} \sin t$$

$$y_p = e^{2t} \frac{1}{D^2 + 1} \sin t$$

Step 4: Solve Trigonometric part (Case of Failure) Apply Trig Rule for $\sin(1t)$ (Replace $D^2 \to -1^2 = -1$): Denominator becomes $-1 + 1 = 0$. This is a Case of Failure. Multiply by t (our new independent variable) and differentiate denominator ($2D$):

$$y_p = e^{2t} \cdot t \cdot \frac{1}{2D} \sin t$$

The operator $\frac{1}{D}$ means integrate with respect to t:

$$y_p = \frac{te^{2t}}{2} \int \sin t \, dt = \frac{te^{2t}}{2}(-\cos t)$$

$$y_p = -\frac{t}{2} e^{2t} \cos t$$

Step 5: Write General Solution in t

$$y(t) = e^{2t}(C_1 \cos t + C_2 \sin t) - \frac{t}{2} e^{2t} \cos t$$

Step 6: Resubstitute back to x Substitute $t = \ln x$ and $e^{2t} = x^2$:

$$y(x) = x^2[C_1 \cos(\ln x) + C_2 \sin(\ln x)] - \frac{1}{2}(\ln x) x^2 \cos(\ln x)$$

Congratulations! You have completed the structured course notes for Ordinary Differential Equations based on your provided material.